

Surgical Basin Portfolio Element A:

Presentation and Justification of the Problem

Project Identification and Definition

During surgeries, surgeons often use special medical-grade basins to hold tools or sterile supplies or drain bodily fluids. In a hospital, it's of the utmost importance to keep the areas around a person's body sterile and sanitized to avoid risking allowing harmful pathogens into someone's open wounds, therefore they require a very specific type and grade of container that is called an emesis or irrigation basin. These specialized bowls are crucial in a hospital setting. However, they aren't perfect. Often when surgeons are trying to drain fluid from an extremity, they need to move the arm or leg to rest on top of the basin, but the shape of the basin can make it difficult to properly drain the fluid, and it can cause discomfort or bruising for some patients. An adapted model of this basin could solve this issue.



Problem Elaboration

If designed correctly, the basin could aid surgeons in the draining process and help to protect and comfort patients, and surgeons across the globe could take advantage of its use. As of right now, the design's rigidity and lack of adaptability is causing doctors to struggle with some patient's unwieldy extremities. The current basin makes it easy for spills to happen during operations, creating a more difficult situation for the surgeons, first assists, and scrub techs in the operating room. A more ergonomic, adaptable, surgery-compatible basin would solve the problem. Creating a basin that is the exact same price (or cheaper) as the typical basin could widely address complications faced by medical professionals across the globe. If everyone who needs one can afford it, then the basin can reach beyond the operating rooms and into communities that need medical assistance for emergency surgeries.

Stakeholder Concerns

Dr. Amir Behnam is our primary stakeholder and the person who came to us with his issue. As a plastic surgeon, he often encounters problems with his current irrigation basin, especially during wound irrigation and debridement. He notices that during the process of irrigation, you often have to rest the patient's arm or leg on top of the basin, and that can cause bruising for patients. It's awkward and difficult to handle extremities while they are resting on the basin because the basin does not form-fit to the arms and legs of the patient.

Other stakeholders include:

- Scrub technicians and nurses: Must manage spills and maintain sterility, adding to workload and increasing risk of infections with patients with exposed wounds.
- Patients: Experience discomfort, bruising, or prolonged recovery from poor limb support.
- Hospital administrators: Concerned with maintaining safety standards, reducing procedure time, and keeping costs low.

Problem Justification

Clinical experience and daily operating room practices justify this issue: surgeons and operating room personnel report challenges with improper limb positioning and fluid spillage. These challenges can slow down procedures, create more chances for contamination, and cause unnecessary discomfort for patients. Current basins aren't designed to support limbs or irrigate fluid efficiently, forcing staff to constantly readjust positions and manage spills during surgery, according to Drexel medical students and Dr. Behnam. There is a large demand for innovative basins to support limb systems in order to make further advancements to modern medicine.

Improving the basin's design would address these concerns. This product would provide stable limb support, improved fluid control, and a more ergonomic shape that can comfortably accommodate different extremities. It would allow surgeons to work more efficiently, reduce the chance of spillage, and increase the comfortability for patients during the procedure. Not only would this improve surgical experiences with surgeons and patients, it would facilitate quicker surgeries in order for surgeons to address more patients throughout the day, saving more lives. The need for a basin like this is very important because many patients and surgeons complain about all issues addressed above extremely often, and this would help significantly.